READ.ME

📌 Project Overview

This project focuses on analyzing healthcare data to understand **patient readmission patterns and hospital resource utilization**.

The dataset contains **10,000 patient records and 26 variables**, including patient demographics, hospital information, doctor details, waiting time, length of stay, discharge status, diseases, and readmission status.

The main objective is to identify factors associated with patient readmission and generate useful insights that can support better healthcare management and hospital resource planning.

🎯 Objectives

The main objectives of this project are:

- Analyze patient demographics such as age, gender, and race.
- Study patient readmission patterns.
- Examine the relationship between length of stay and readmission.
- Analyze hospital bed availability and occupancy.
- Evaluate patient waiting times.
- Study patient satisfaction scores.
- Identify common diseases among patients.
- Analyze discharge status.
- Explore hospital wards and department referrals.
- Identify factors that may be associated with higher readmission rates.

📂 Dataset Information

Dataset: `Healthcare Data Analysis for readmission.csv`

Number of Records: 10,000
Number of Columns: 26

Dataset Columns

Column	Description
`hospital_name`	Name of the hospital
`Admission_date`	Patient admission date
`hospital_id`	Unique hospital identifier

Column	Description
`hospital_beds_available`	Number of beds available in the hospital
`occupied_beds`	Number of occupied hospital beds
`hospital_ward`	Ward where the patient was admitted
`patient_id`	Unique patient identifier
`patient_gender`	Gender of the patient
`patient_age`	Age of the patient
`patient_race`	Race/ethnicity category
`patient_sat_score`	Patient satisfaction score
`patient_first_initial`	First-name initial
`patient_last_name`	Patient last name
`patient_waittime`	Patient waiting time
`department_referral`	Department to which the patient was referred
`time_slot`	Patient appointment/admission time slot
`doctor_id`	Unique doctor identifier
`doctor_name`	Name of the doctor
`doctor_specialty`	Doctor's specialty
`patient_assigned_doctor`	Whether a doctor was assigned
`patient_checkin_date`	Patient check-in date
`patient_checkout_date`	Patient check-out date
`patient_disease`	Disease/medical condition
`patient_length_of_stay`	Length of hospital stay
`discharge_status`	Patient discharge status
`readmission`	Patient readmission indicator

🔑 Target Variable

The primary target variable is:

```
readmission
```

- **0** → Patient was not readmitted
- **1** → Patient was readmitted

This variable is used to study the factors that may influence hospital readmission.

🧹 Data Cleaning

The data-cleaning process can include:

- Checking missing values
- Checking duplicate records
- Correcting inconsistent data types
- Converting date columns to datetime format
- Checking categorical values
- Detecting unusual or invalid values
- Removing unnecessary columns when required
- Preparing the dataset for visualization and analysis

📊 Exploratory Data Analysis

The project explores several important healthcare questions, including:

1. What is the gender distribution of patients?
2. Which age groups have the highest number of patients?
3. What is the patient readmission distribution?
4. Does patient age affect readmission?
5. Is length of stay related to readmission?
6. Which diseases are most common?
7. Which hospital wards receive the most patients?
8. How does waiting time vary among patients?
9. What is the relationship between discharge status and readmission?
10. How are hospital beds being utilized?

📈 Example Visualizations

Visualizations used in the analysis may include:

- Bar charts
- Count plots
- Histograms
- Box plots
- Pie charts
- Correlation heatmaps
- Readmission comparison charts
- Age-group analysis
- Length-of-stay analysis
- Hospital resource utilization charts

💡 Key Analysis Areas

Patient Demographics

Patient age, gender, and race can be analyzed to understand the demographic characteristics of the hospital population.

Patient Readmission

The `readmission` variable can be compared with age, disease, hospital ward, discharge status, waiting time, and length of stay.

Length of Stay

Comparing `patient_length_of_stay` between readmitted and non-readmitted patients can help determine whether hospitalization duration is associated with readmission.

Hospital Resource Utilization

Available and occupied beds can be analyzed to understand hospital capacity and resource utilization.

Patient Experience

Waiting time and satisfaction scores can provide information about patient experience and hospital service efficiency.

🛠️ Technologies Used

- Python
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

📚 Python Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

🚀 How to Run the Project

1. Download or clone this repository.
2. Open the notebook in **Google Colab** or **Jupyter Notebook**.
3. Upload the healthcare CSV dataset.
4. Install/import the required Python libraries.
5. Run the data-cleaning cells.
6. Perform exploratory data analysis.
7. Generate visualizations.
8. Interpret the results and prepare conclusions.

📌 Conclusion

This healthcare data analysis project provides insights into **patient readmission, patient demographics, hospital stays, waiting times, diseases, discharge outcomes, and hospital resource utilization**.

By analyzing these factors, hospitals can better understand patterns associated with patient readmission and identify opportunities to improve resource planning, patient care, and operational efficiency.

👩‍💻 Project Type

Healthcare Analytics – Patient Readmission Patterns and Hospital Resource Optimization

📄 License

This project is intended for educational and data-analysis purposes.